Microcontroller Based Application Development Project – IN 1901

Project Proposal Report

LEVEL 01

Smart Laser-Based Home Security System with IoT Alerts and AI Monitoring

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Introduction

Security is a major concern in modern society, especially when homes are left unattended. This project introduces a smart laser-based home security system that detects intrusions using sensors and laser technology. When the laser beam is interrupted, a notification is instantly sent to the user's mobile device via IoT technology. The system then captures an image and transmits it to the homeowner's phone. After receiving the image, the system can identify the intruder and trigger the necessary alarms.

Aim and Objectives

The aim of this project is to design a smart laser-based security system for home protection. It detects intrusions, sends real-time alerts, captures images, and integrates hardware with software to ensure reliability, cost-effectiveness, and suitability for modern security applications.

Problem and Proposed Solution

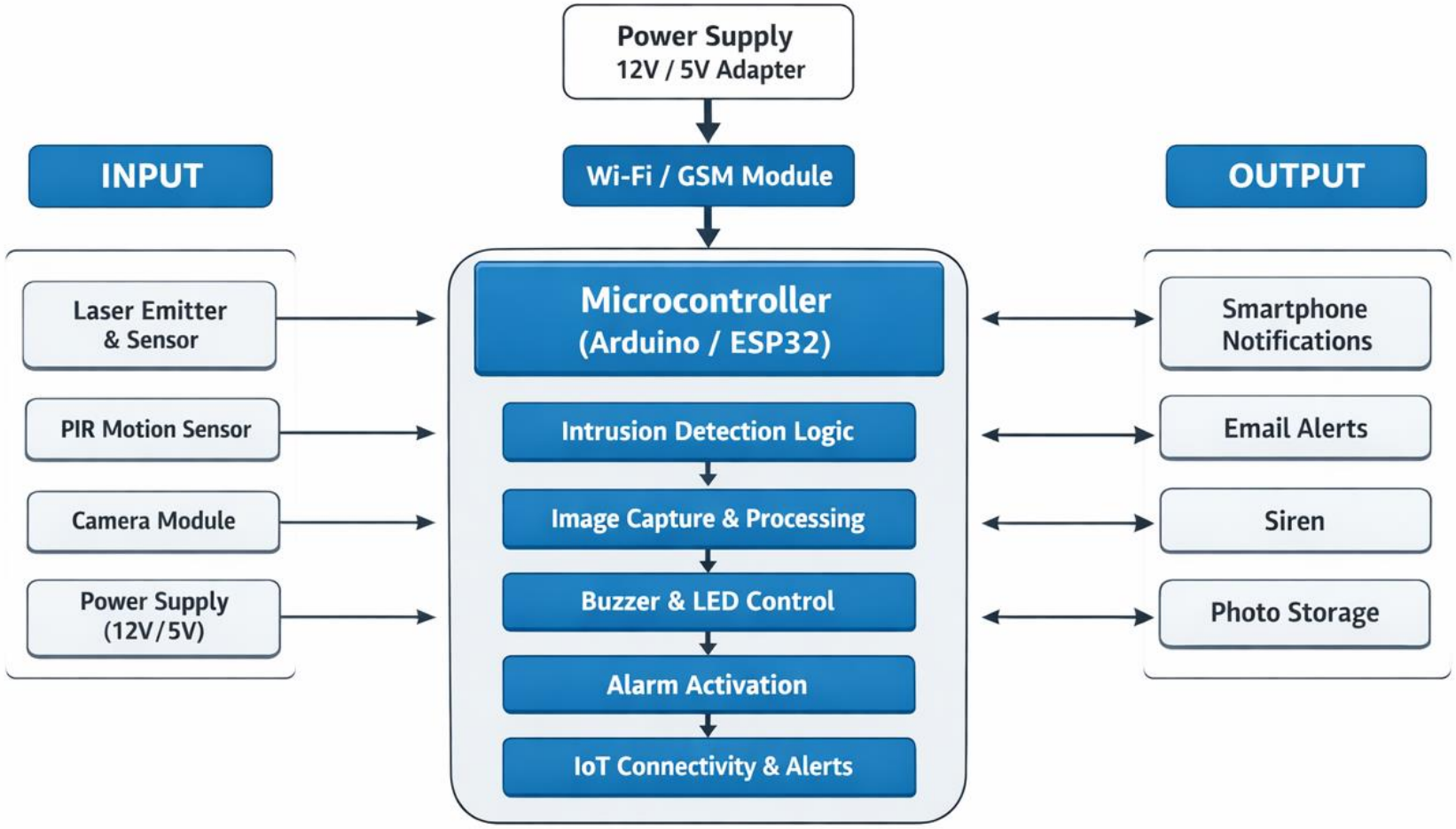
Problem:

Home security is a major concern, especially when owners are away. Traditional systems only provide alarms without real-time monitoring or remote notifications, making them less effective against intrusions.

Proposed Solution : 💡

This project proposes a smart laser-based security system that detects intrusions, sends real-time alerts, captures images, and integrates IoT technology to provide reliable and efficient home protection.

Block Diagram



Components used and Technologies

Microcontroller
Arduino Uno / ESP32

Communication
ESP8266 / ESP32 Wi-Fi
HTTP / MQTT Protocol

Sensors
LDR (Light Dependent Resistor)
Laser Module
PIR Motion Sensor (optional)

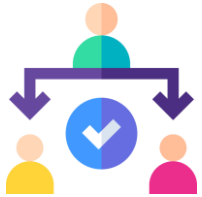
Technologies
Embedded C Programming
IoT Systems
Image Capture & Processing
Mobile Notification (Blynk / Firebase / Telegram)

Actuators
Buzzer (Alarm)
LED Indicator
Camera Module (ESP32-CAM)

Expected Budget

Component	Estimated Cost (LKR)
ESP32 Microcontroller	1,800
ESP32-CAM Module	2,500
Laser Module + LDR	500
PIR Motion Sensor	800
RFID Module / Keypad	1,000
Alarm Sound (Buzzer/Speaker)	500
Power Supply	1,500
Wires, PCB, Enclosure	5,000
Miscellaneous	2,400
Total Cost	**16,000**

Work Distribution



Member	Responsibility Area	Tasks
Member 1	Laser & Sensor System	Design laser detection, connect LDR/PIR sensors, calibration, testing
Member 2	Microcontroller & Hardware	Program ESP32/Arduino, connect components, system integration
Member 3	IoT & Communication	Develop WiFi alerts, mobile notifications, cloud/database handling
Member 4	Camera & Image Processing	Setup ESP32-CAM, capture images, send images to mobile
Member 5	UI & Security Control	Develop app/dashboard, implement RFID/keypad, system control

References

No	Source	Description	Link
1	Espressif Systems	ESP32 Datasheet and Technical Reference Manual	https://www.espressif.com
2	Arduino	Arduino Programming and Hardware Documentation	https://www.arduino.cc
3	Random Nerd Tutorials	ESP32-CAM and IoT Projects Guide	https://randomnerdtutorials.com
4	Electronics Hub	Laser Security System Using Arduino	https://www.electronicshub.org
5	Google Firebase	Realtime Database and Cloud Messaging Documentation	https://firebase.google.com
6	OpenCV	Open Source Computer Vision Library	https://opencv.org

THANK YOU … !